

DERIVATE DI ORDINE SUCCESSIVO

• $f(x) = \arctan x$

$$f'(x) = \frac{1}{x^2 + 1} \quad \leftarrow \text{DERIVATA PRIMA}$$

$$f''(x) = f'(x)' = -\frac{1}{(x^2 + 1)^2} \cdot 2x$$

• $f(x) = 3x^2 + 4x$

$$f'(x) = 6x + 4 \quad f''(x) = 6$$

DEF. I INTERVALLO, $f: I \rightarrow \mathbb{R}$

SI DICE DERIVABILE 2 VOLTE SE:

a) f É DERIVABILE

b) f' É DERIVABILE

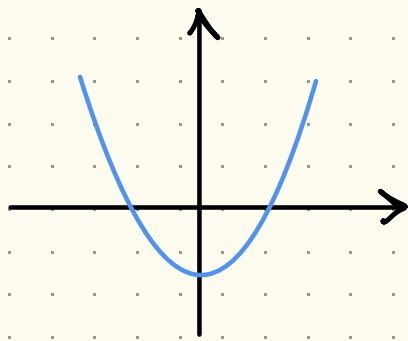
LA FUNZIONE $f'': I \rightarrow \mathbb{R} \quad f'' = (f')'$

SI CHIAMA DERIVATA SECONDA.

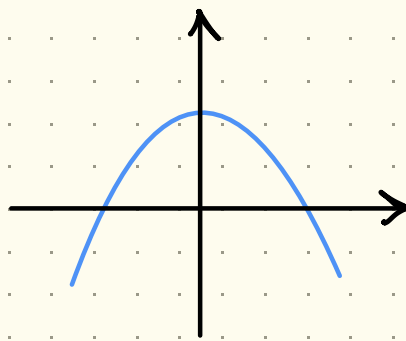
ANALOGAMENTE

$$\text{DERIVATA } m\text{-ESIMA} \rightarrow f^{(m)} = (f^{(m-1)})' = f^{''''\dots'} \quad \leftarrow m \text{ VOLTE}$$

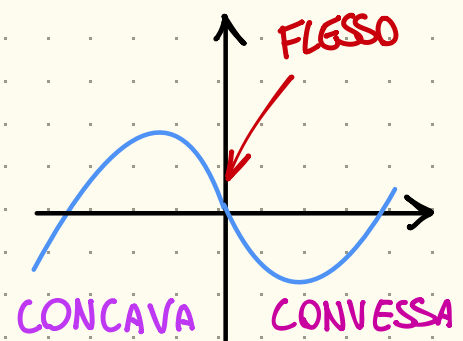
CONCAVE e CONVESSE



CONVESSA



CONCAVA

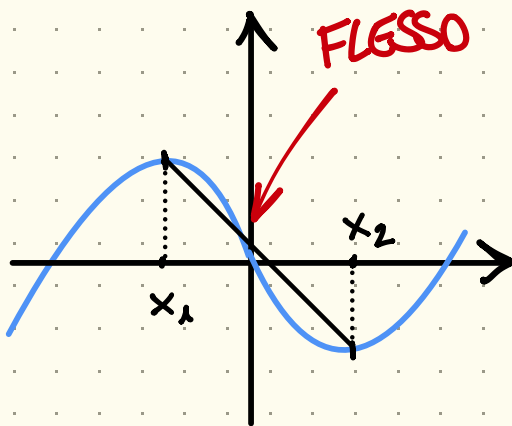


DEF. $f: I \rightarrow \mathbb{R}$ É DETA

CONVESSA
CONCAVA

SE $\forall x_1, x, x_2 \in I, x_1 \leq x \leq x_2$

$$f(x) \begin{matrix} \leq \\ \geq \end{matrix} f(x_1) + \frac{x - x_1}{x_2 - x_1} (f(x_2) - f(x_1))$$



NÉ CONCAVA, NÉ CONVESSA

DEF. $f: I \rightarrow \mathbb{R}$ a ESTREMO SX DI I
 b ESTREMO DX DI I

UN PUNTO $a < x_0 < b$ SI DICE PUNTO DI FLESSO SE $\exists \delta > 0$ T.C.

- f CONVESSA IN $[x_0 - \delta, x_0]$
CONCAVA IN $[x_0, x_0 + \delta]$

OPPURE

- f CONCAVA IN $[x_0 - \delta, x_0]$
CONVESSA IN $[x_0, x_0 + \delta]$

TEOREMA

$f: I \rightarrow \mathbb{R}$ DERIVABILE 2 VOLTE

SONO FATTI EQUIVALENTI:

a) f È CONVESSA
CONCAVA

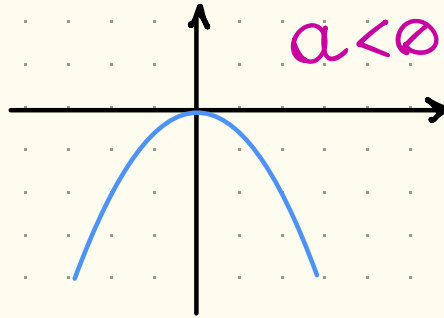
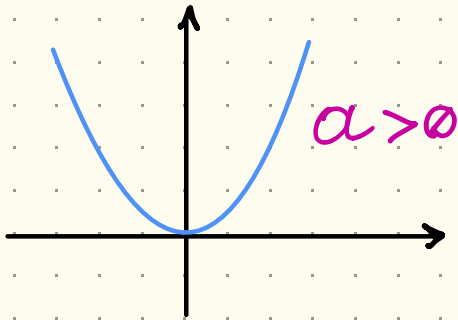
b) $\forall x_0, x \in I$

$$f(x) \begin{matrix} \geq \\ \leq \end{matrix} f(x_0) + f'(x_0)(x - x_0)$$

$$c) \forall x \in I \quad f''(x) \begin{matrix} \geq \\ \leq \end{matrix} 0$$

ESEMPI

$$f(x) = ax^2 \quad a \neq 0$$



$$f'(x) = 2ax$$

$$f''(x) = 2a$$

$a > 0 \Rightarrow f''(x) > 0 \Rightarrow f$ CONVESSA
 $a < 0 \Rightarrow f''(x) < 0 \Rightarrow f$ CONCAVA

es)

$$\bullet f(x) = x^2 + \sin x$$

$$f'(x) = 2x + \cos x$$

$$f''(x) = 2 - \sin x \geq 1 > 0 \quad \forall x \in \mathbb{R}$$

$\Rightarrow f$ CONVESSA $\Rightarrow \nexists$ FLESSI

$$\bullet f(x) = e^x$$

$$f'(x) = e^x > 0 \quad \forall x \in \mathbb{R} \Rightarrow f. \text{ CRESC.}$$

$$f''(x) = e^x > 0 \quad \forall x \in \mathbb{R} \Rightarrow f. \text{ CONVESSA}$$

$$f(0) = 1 \quad ; \quad f'(0) = 1$$

tan IN 0: $y = 1 + 1 \cdot x = x + 1$

$$\bullet f: (-1, +\infty) \rightarrow \mathbb{R}, f(x) = \log(x+1)$$

$$f'(x) = \frac{1}{x+1} \cdot (x+1)' = \frac{1}{x+1} \Rightarrow f'(x) > 0$$

$$\forall x > -1$$

$$f'(x) = \frac{1}{(x+1)^2} \cdot (x+1)' = -\frac{1}{(x+1)^2} < 0 \quad \forall x$$

$$\Downarrow \\ f. \text{ CRESC.}$$

$$\Downarrow \\ f. \text{ CONCAVA}$$

es)

Studiare la convessità della funzione

$$f(x) = \frac{x^3}{3} - x^2 - 3x + 1.$$

$$f(x) = \frac{x^3}{3} - x^2 - 3x + 1$$

$$f'(x) = \frac{1}{3} \cdot 3x^2 - 2x - 3$$

$$f''(x) = \frac{1}{\cancel{3}^1} \cdot \cancel{6}^2 x - 2 = 2x - 2$$

$$2x - 2 > 0$$

$$2x > 2$$

$$x > 1$$

$$\begin{array}{c} 1 \\ - | + \\ \cap \cup \end{array} \quad \begin{array}{l} (-\infty, 1] \text{ CONCAVO} \\ (-1, +\infty) \text{ CONVESSO} \end{array}$$

